

ezDAQ – User Manual

Data acquisition, live view and analysis with NI cDAQ

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1 Introduction

ezDAQ is a Windows application for data acquisition with National Instruments measurement systems (NI cDAQ). It covers the whole course of a measurement:

- **Configuration:** detect connected modules, assign, name and scale channels, set the sample rate and storage options,
- **Measuring:** display readings live and write them to disk at the same time, started and stopped manually or by a trigger,
- **Analysis:** load stored measurements, plot and inspect them and evaluate them with FFT, filters and smoothing.

This manual explains how to use ezDAQ step by step. To get going quickly, start with chapter Quick start: your first measurement.

1.1 Supported modules

Module	Measurement	Channels	Sample rate
NI 9215	Voltage ±10 V	4	freely selectable
NI 9234	Voltage ±5 V or IEPE (accelerometer, microphone)	4	only 51,200 Hz / n (n = 1 … 31), i.e. 51,200 … 1,651.6 S/s
NI 9210	Thermocouple	4	at most 14.3 S/s
NI 9213	Thermocouple (J, K, T, E, N, R, S, B)	16	at most 100 S/s, depending on channel count and timing mode
NI 9235	Strain gauge, 120 Ω quarter bridge	8	only 50,000 Hz / n (n = 5 … 63), i.e. 10,000 … 793.7 S/s

Modules of different types can be combined in one measurement. If they cannot share a common clock (for example an NI 9210 next to a fast NI 9234), ezDAQ automatically splits the measurement into several groups that run in sync. See section Sample rate.

1.2 Requirements and installation

- **Windows 10 or 11.**
- **NI-DAQmx driver (runtime)** from National Instruments. The driver is not part of ezDAQ and has to be installed once, with administrator rights. Without it ezDAQ still starts, but finds no devices.
- **ezDAQ itself** comes as an installer (`ezDAQ-Setup-<version>.exe`) or as a folder that is started directly (`ezDAQ.exe`). At the start, the installer asks whether to install for all users (with administrator rights) or only for yourself (without administrator rights).

> **Note:** On the first start, Windows SmartScreen may warn about an “unknown publisher” because the program is not signed. Use *More info* → *Run anyway* to start it.

ezDAQ’s settings (last configuration, language, theme, window positions) are stored in your user profile under `%APPDATA%\ezDAQ`. Measurement data goes to the folder you choose as the storage location.

1.3 The user interface at a glance

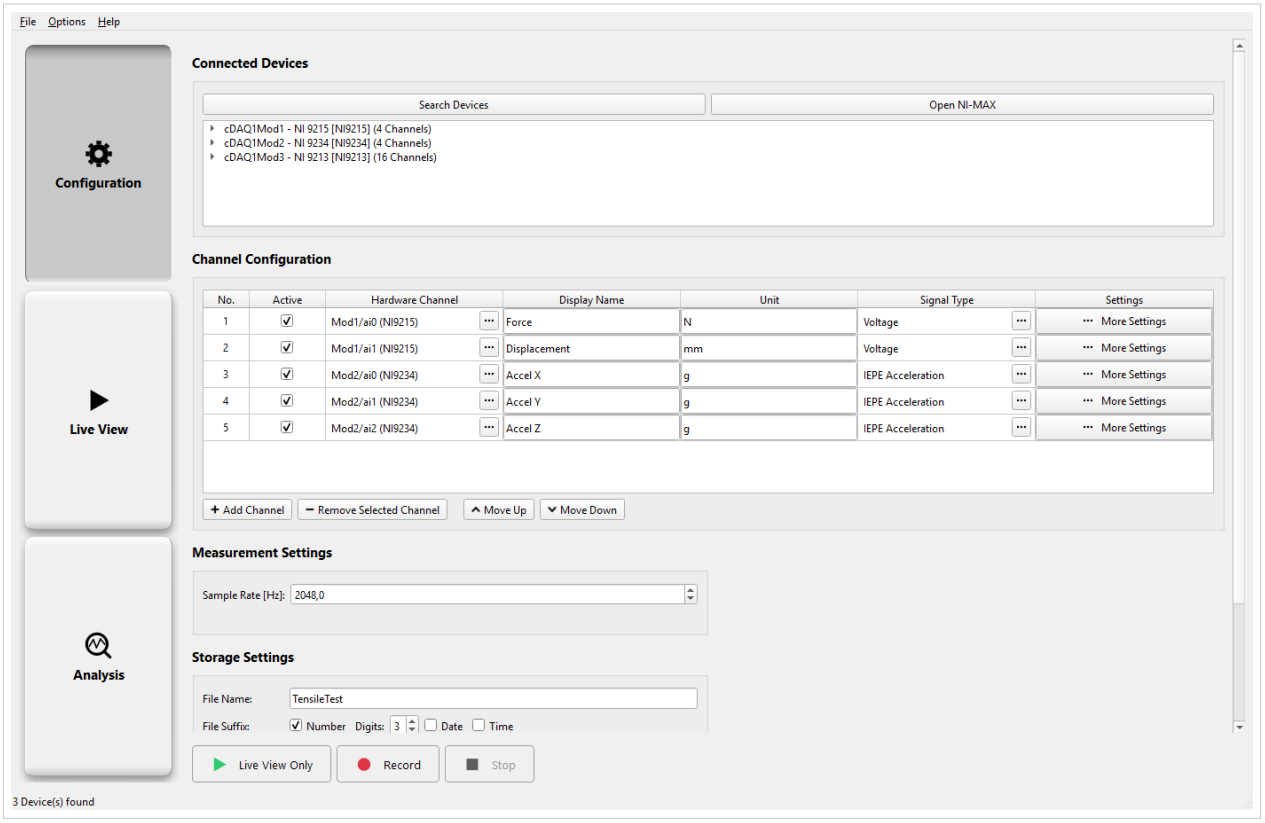


Figure 1.1: The main window with the configuration page

- **Navigation (left):** *Configuration*, *Live View* and *Analysis*. A click switches the page; a running measurement keeps running.
- **Menu bar:**
  - *File:* save and load configuration, load measurement, quit.
  - *Options:* language, theme (light/dark), live view display, sensor database, trigger settings.
  - *Help:* this manual (**F1**), check for updates, about.
- **Status bar (bottom):** shows what is happening, e.g. how many devices were found or whether a measurement is running. While data is being stored, a **red dot** is shown on the left.

2 Quick start: your first measurement

1. **Connect the hardware** and start ezDAQ. Devices are searched for automatically at start-up and appear under *Connected Devices*.
2. **Add a channel:** under *Channel Configuration*, click **Add Channel**. ezDAQ assigns the first free hardware channel to the new row.
3. **Set up the channel:** enter display name and unit; if needed, set signal type, scaling or sensitivity via **Signal Type** and **More Settings**.
4. Enter the **sample rate** under *Measurement Settings*.
5. **Set up storage:** under *Storage Settings*, enter a file name and choose the target folder with **Choose Storage Location**.
6. **Test without saving: Live View Only** shows the readings without storing anything. Use it to check wiring and scaling.
7. **Record: Record** (or **F5**) starts the measurement with storage, **Stop** (or **F8**) ends it.
8. **Evaluate:** on the *Analysis* page, drop the stored file in and drag a channel onto the plot.

> **Tip:** *File → Save Configuration* stores a finished setup as a file; *Load Configuration* restores it later.

3 Configuration

The *Configuration* page holds everything that is set before a measurement. ezDAQ remembers the last state and restores it on the next start.

3.1 Connected devices

The list shows every module reported by the NI-DAQmx driver, with its type and channel count. Click the arrow in front of a module to expand its channels.

- **Search Devices** searches again, e.g. after plugging in a module. The list is also refreshed automatically when you switch to the configuration page.
- **Open NI-MAX** starts the *NI Measurement & Automation Explorer*, where devices can be reserved, renamed and tested.

Two special cases are marked in the list and reported with a message:

- **not connected:** the device is known to the driver but does not respond, e.g. an unplugged or unreachable network cDAQ chassis.
- **not supported:** ezDAQ does not support this module (yet).

The channels of such devices cannot be selected.

3.2 Channel configuration

Each row of the table is one measurement channel:

Column	Meaning
No.	running number, also the column order in the measurement file
Active	only active channels are measured and stored
Hardware Channel	the physical input, e.g. `Mod1/ai0 (NI9215)`
Display Name	name of the channel in the live view, measurement file and analysis
Unit	physical unit of the scaled value, e.g. N, mm, g, °C
Signal Type	voltage, IEPE acceleration, thermocouple or strain
Settings	**More Settings:** scaling, sensitivity etc. (see below)

Below the table:

- **Add Channel** adds a row with the first free hardware channel.
- **Remove Selected Channel** deletes the selected rows.
- **Move Up / Move Down** move the selected rows, with all their settings. The order is the column order in the measurement file. Hold Ctrl to select several rows and move them together.

3.2.1 Assigning and swapping hardware channels

Clicking the **...** button in the *Hardware Channel* column opens the channel picker, grouped by module.

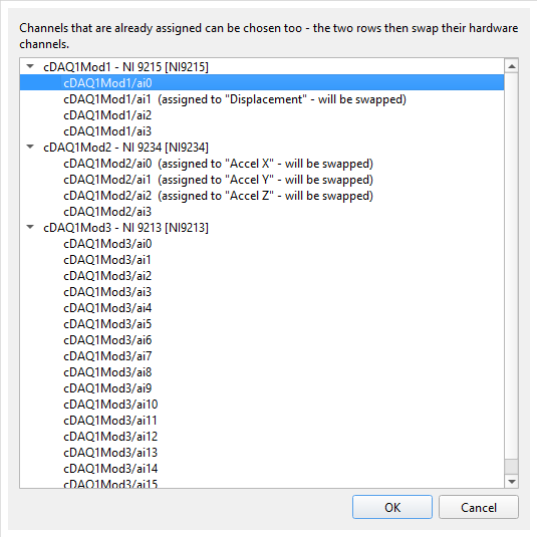


Figure 3.1: Channel picker with channels already assigned

Channels already assigned to another row are listed with *assigned to "…" – will be swapped*. Choosing such a channel makes **the two rows swap their hardware channels**. Name, unit and all other settings stay in their row. This way the assignment can be changed even when every input is in use.

If the new channel is on a different module type, ezDAQ adjusts the signal type when the new module cannot do the old one. An IEPE channel moved to an NI 9215, for example, becomes a voltage channel.

3.2.2 Signal type

The available signal types depend on the module:

Module	Signal types
NI 9215	Voltage
NI 9234	Voltage, IEPE acceleration
NI 9210, NI 9213	Thermocouple
NI 9235	Strain

With **IEPE acceleration**, the NI 9234 powers the sensor with the IEPE constant current (2 mA). The driver converts the voltage to g directly, using the sensitivity entered.

3.2.3 More Settings

More Settings opens the parameters of the channel. Only the fields that matter for the signal type are shown.

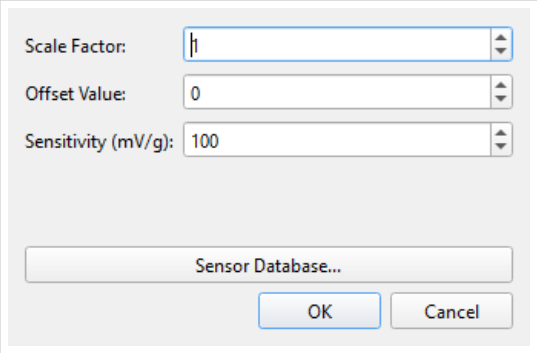


Figure 3.2: Settings of an IEPE channel

- **Scale Factor** and **Offset Value** (all signal types): ezDAQ converts every reading as

  value = raw value × scale factor + offset.

  Example: a force sensor outputs 0 … 10 V for 0 … 2,500 N, so scale factor 250, offset 0, unit N.
- **Sensitivity (mV/g)** (IEPE): from the sensor's calibration certificate.
- **Thermocouple type** (thermocouple): J, K, T, E, N, R, S or B.
- **ADC timing mode** (NI 9213 only): *High Resolution* (more accurate, slower) or *High Speed*. It always applies to all channels of the same module and limits the possible sample rate (see Sample rate).
- **Gage Factor (k)**, **Bridge Type** and **Lead Wire Resistance (Ω)** (strain): data of the strain gauge. Bridge type *Quarter Bridge (1 active gage)* or *Quarter Bridge (1 active + 1 dummy gage)*.
- **2-Point Calibration …** (thermocouple): enter two known reference points, e.g. ice water (0 °C) and boiling water (100 °C), each with the measured and the known value. ezDAQ calculates scaling and offset from them.
- **Sensor Database …** opens the sensor catalog to look up sensitivities (see Sensor database).

3.3 Measurement settings

3.3.1 Sample rate

The **sample rate [Hz]** is the number of readings per second and channel. Some modules cannot do every rate:

- **NI 9234:** only 51,200 Hz / n (n = 1 … 31). At the start, ezDAQ reports the next valid value, e.g. 2048 Hz instead of 2000 Hz.
- **NI 9235:** only 50,000 Hz / n (n = 5 … 63).
- **NI 9210:** at most 14.3 S/s.
- **NI 9213:** at most 100 S/s for the whole module. In *High Resolution* mode all active channels of a module share about 18 S/s, i.e. about 1.1 S/s with 16 channels; in *High Speed* mode it is about 84 S/s with 16 channels.

Below the input field, ezDAQ shows the rate that is actually used. If the requested rate exceeds the limit of a module, that module measures at its maximum and the others keep the target rate. In the measurement file all channels then share one time base: the slower channels repeat their last value until a new one arrives, and their actual rate is recorded in the metadata.

3.4 Storage settings

- **File name:** base name of the measurement, e.g. `TensileTest`.

- **File suffix:** appended automatically, in the order *Name_Date_Time_Number*:
  - **Number** with a selectable number of digits (`_001`, `_002`, …): the next free number in the storage location is used.
  - **Date** (YYYYMMDD) and **Time** (HHMMSS) of the start.
- **Storage format: Parquet (recommended)** or **CSV**, see Measurement files.
- **Recording Limit / Unlimited:** with *Unlimited (until disk is full)* unchecked, every recording ends automatically after the set number of *samples*, *seconds*, *minutes* or *hours*. This is counted on the clock of the measurement hardware, not the PC clock.
- **Storage location:** folder the measurement files are written to.

> **Important:** ezDAQ never overwrites an existing measurement. With the number suffix switched off and the name already taken in the storage location, an error message is shown and the measurement does not start.

3.5 Saving and loading a configuration

File → Save Configuration … stores the complete setup as a JSON file: channels with all settings, sample rate, storage options and the live view display. *File → Load Configuration …* restores it. This makes recurring tests quick to set up and easy to hand over to colleagues.

3.6 Sensor database

Options → Sensor Database … (or the button in the channel settings) holds a catalog of the available sensors: category, manufacturer, serial number, notes and, per measurement axis, one or more range variants with their sensitivity.

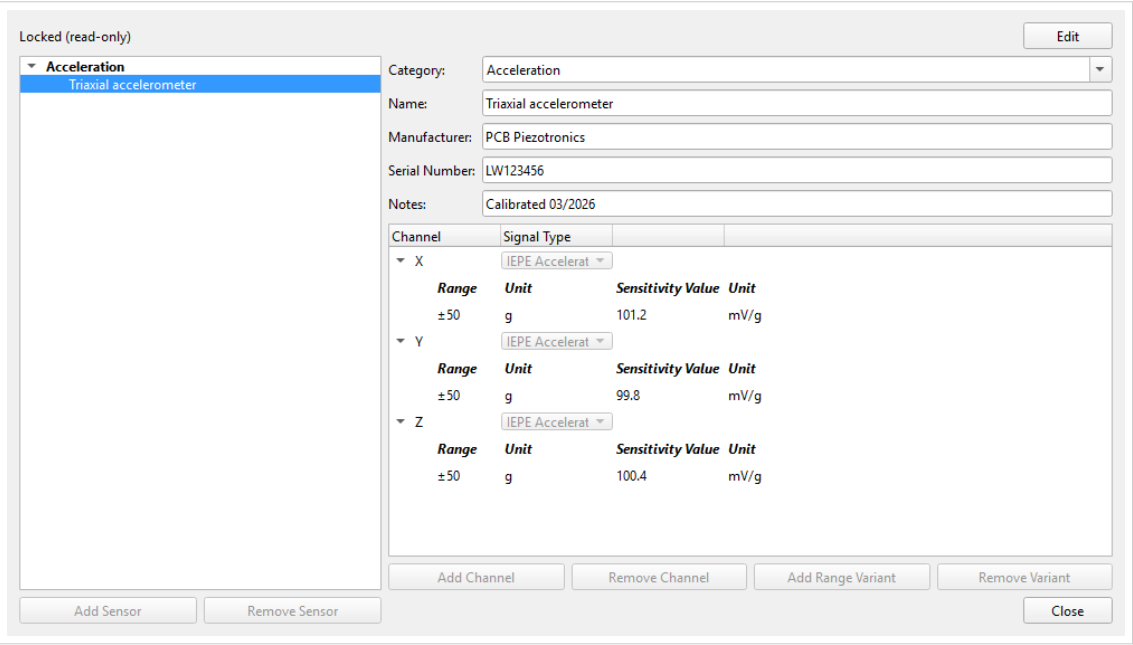


Figure 3.3: Sensor database

The database is for looking things up; the matching sensitivity is entered into the channel settings by hand. It is write-protected: click **Edit** and enter the password (ask your lab supervisor) to make changes. Changes are saved immediately.

4 Measuring

4.1 Starting and stopping

The three buttons are available on the configuration page and in the live view:

Button	Key	Effect
Live View Only		measures and displays, **but stores nothing**
Record	F5	measures, displays and **stores** to the storage location
Stop	F8	ends the measurement; the file is finalized

Before starting, ezDAQ checks the settings: whether every active channel has a hardware channel, whether the sample rate suits the modules and whether a storage location is set. Any problem is explained in a message.

During a measurement the status bar shows whether data is stored (*Recording: … – data is being saved*, with a red dot) or only displayed (*Live view: … – nothing is being saved*).

4.2 Storage buffer

During a recording, the bottom of the live view shows the **Storage Buffer (Write Backlog)**: the current file size and how many readings have not been written to disk yet. Normally the bar stays near 0 % and is green; from 70 % it turns orange, from 90 % red. If it fills up, readings are lost; ezDAQ reports this at the end of the measurement. The cause is usually a slow or almost full target drive, for example a network drive.

4.3 Trigger

A trigger starts or stops the recording automatically. It is set under *Options → Trigger Settings …*, separately for **Start** and **Stop**.

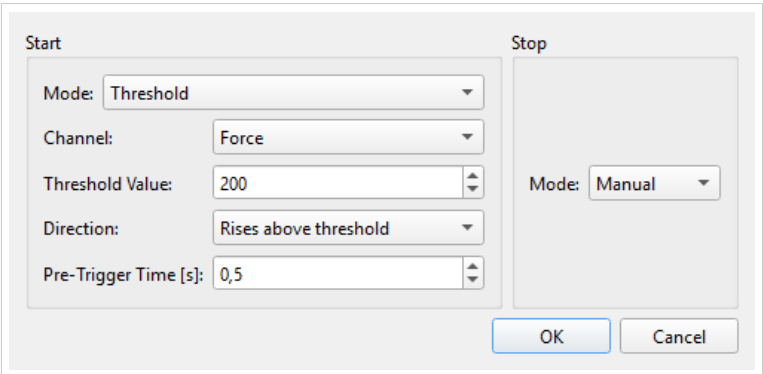


Figure 4.1: Trigger settings with a threshold start

Each side has three modes:

- **Manual:** no trigger; start or stop by hand.

- **Threshold:** fires when a channel crosses a threshold. Set **Channel**, **Threshold** (in the channel's unit) and **Direction**: *Rises above threshold*, *Falls below threshold* or *Magnitude exceeds threshold*. For the start trigger, the **Pre-Trigger Time [s]** sets how many seconds **before** the trigger are stored as well.
- **Serial (USB):** fires when a specific text arrives on a COM port, e.g. from an Arduino or a testing machine. Set **COM port** (re-read with **Refresh**), **Baud rate** and **Expected signal**; only an exact match counts. **Test Serial Trigger** checks beforehand whether the signal arrives.

4.3.1 Measuring with a trigger

1. Set up the trigger and confirm with **OK**.
2. Click **Arm Trigger** (configuration page or live view). The measurement starts and is displayed, but nothing is stored yet. A banner shows what is being waited for.
3. When the start trigger fires, recording begins, including the pre-trigger.
4. The recording ends with the stop trigger, the recording limit or **Stop**.
5. As long as the trigger is armed, ezDAQ then **automatically waits for the next trigger**, each time with a new file (number suffix). This records test series without intervention.
6. **Disarm Trigger** ends the series.

5 Live view

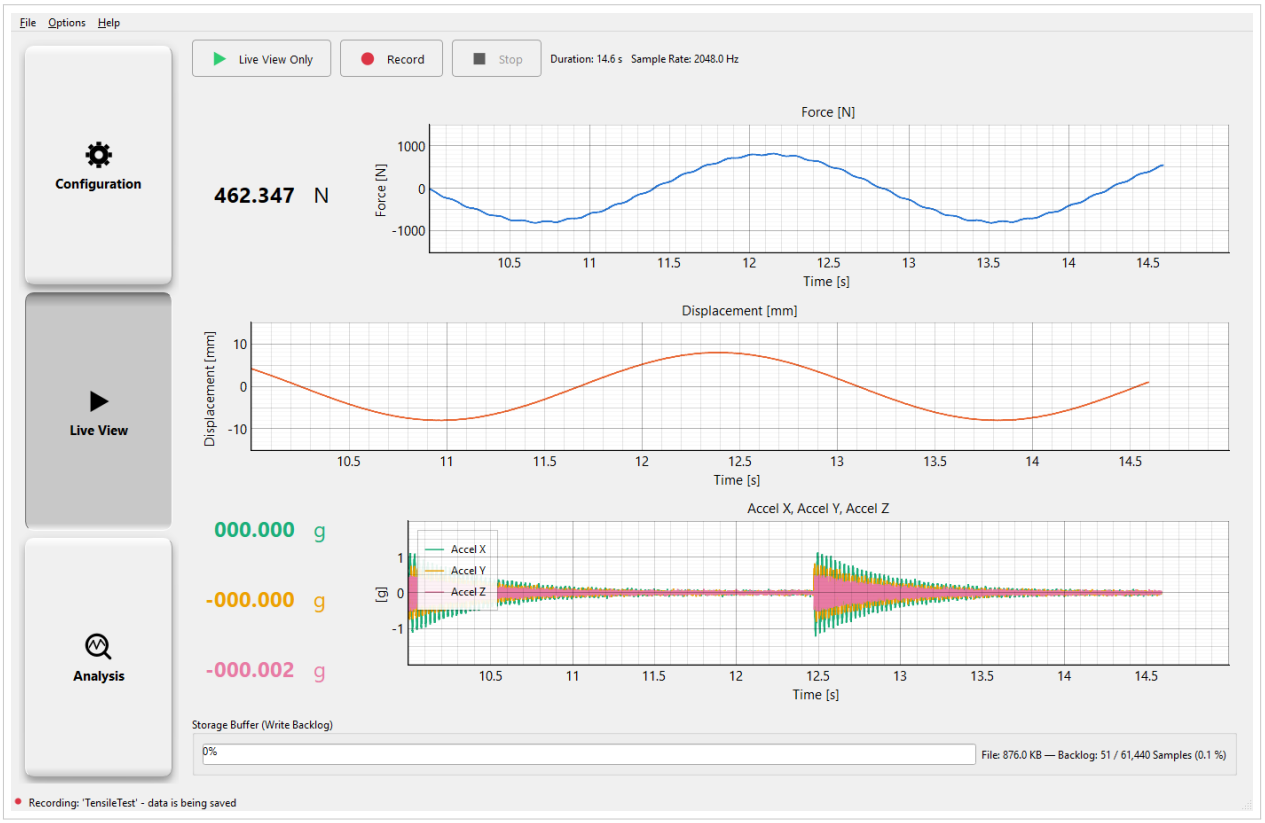


Figure 5.1: Live view during a recording

The live view shows the readings in real time, like an oscilloscope: the curve runs from left to right through the set time window and then starts again on the left. The time axis shows the actual measurement time, e.g. 10 … 15 s in the third pass. Duration and sample rate are shown at the top.

Even before the start, the live view shows the configured channels as empty plots, so the display can be prepared in advance.

5.1 Setting up the display

Options → Set Live View Display opens the live view settings. They apply immediately, also during a running measurement, and are saved with the configuration.



Figure 5.2: Live view display

- **Grid columns:** how many plots are placed side by side (1 … 4).
- Per channel:
  - **Active:** show the channel in the live view. It is measured and stored either way.
  - **Plot group:** channels with the same number share one plot, see Several channels in one plot.
  - **Popout Window:** show the plot in its own, freely movable window instead of in the grid.
  - Checkbox + **Plot:** show the curve; the button opens the plot settings.
  - Checkbox + **Value:** show the current reading in large digits next to the plot; the button opens the value settings.

Plot and value can be combined freely, e.g. only the number without a curve.

5.2 Several channels in one plot

Channels with the **same plot group** are drawn together into one plot; different numbers give separate plots. The default is 1, 2, 3, …, i.e. one plot per channel. The plots are ordered by number, so the numbers also set the order of the plots.

In the picture above, *Accel X*, *Accel Y* and *Accel Z* are in plot group 3 and share one plot:

- The curves keep their colors; a **legend** shows the names.
- There is **one shared Y axis**. With a common unit it is labeled e.g. "[g]"; with different units, all of them are listed. Autoscaling takes all curves into account.
- The values are shown **stacked, in the curve color**.
- Time span, background, grid, Y range, axis titles, *plot on/off* and *popout window* apply to the whole plot: changing them for one channel of the group changes them for all. Curve color, line width and number format stay per channel.

5.3 Plot settings

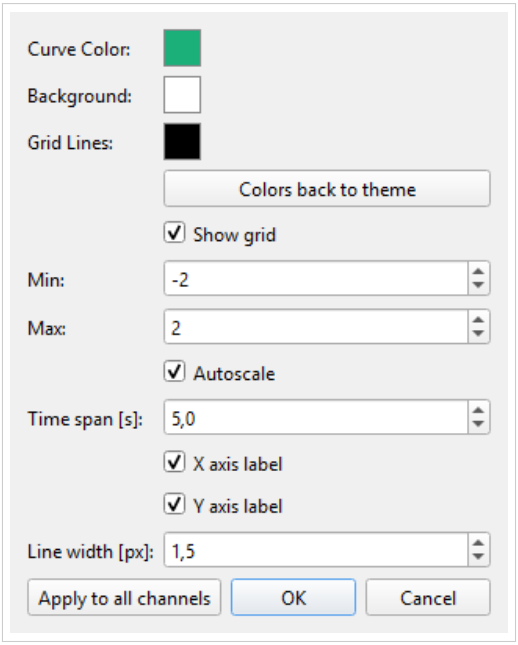


Figure 5.3: Plot settings

- **Curve color, background, grid lines:** click the color swatch. **Colors back to theme** makes them follow the light or dark theme again.
- **Show grid:** grid lines on/off.
- **Min / Max:** fixed Y range.
- **Autoscale:** as long as all values are inside the fixed range, it stays; once a value leaves it, the axis adapts automatically. Unchecked, the fixed range always applies.
- **Time span [s]:** width of the time window.
- **X / Y axis label:** axis titles on/off; saves space with many plots.
- **Line width [px]:** thickness of the curve, worth increasing for projectors or screenshots.
- **Apply to all channels:** copies all values of this dialog to every channel.

5.4 Value

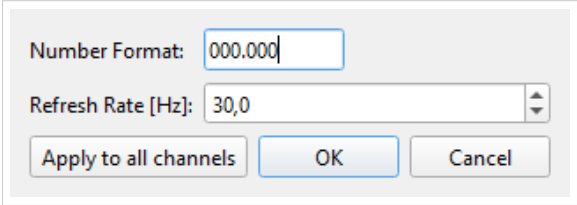


Figure 5.4: Value settings

- **Number Format:** as a pattern, e.g. `000.000` for three integer and three decimal digits. If a value does not fit the integer digits, hash marks (###) are shown instead of a truncated number.
- **Refresh Rate [Hz]:** how often the number is rewritten. In between, readings are **averaged**: the display shows the mean over the interval, not a random single reading. Lower values calm down a noisy display. The curve and the stored data are not affected.

5.5 Popout window

With **Popout Window** a plot – including a shared plot of several channels – appears in a window of its own, e.g. on a second screen. Position and size are remembered. Closing the window brings the plot back into the live view grid.

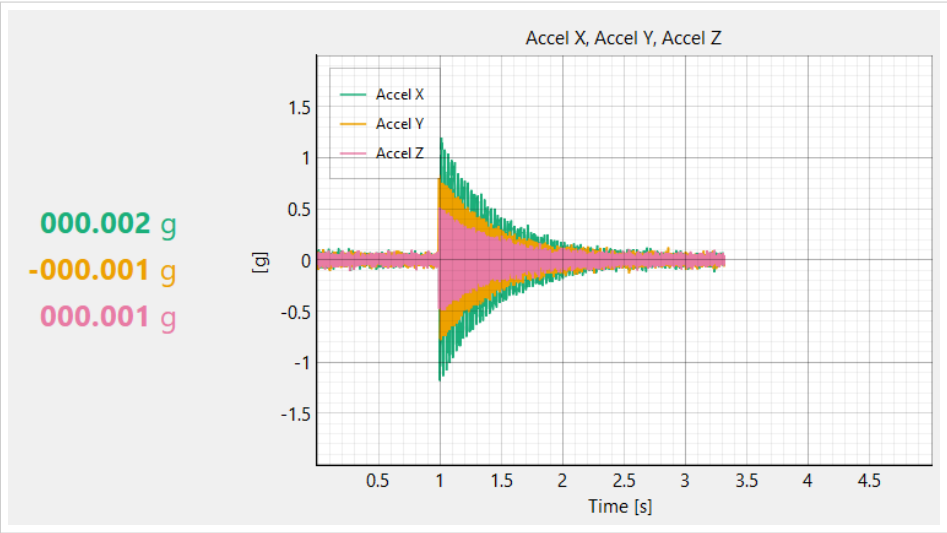


Figure 5.5: Shared plot in a popout window

6 Measurement files

6.1 What is stored

Every recording creates two files in the storage location:

- **`<name>.parquet`** or **`<name>.csv`:** the measurement data,
- **`<name>_info.json`:** the metadata: start time, duration, sample rate(s), hardware and all channels with unit, scaling and settings.

The measurement file is a table with the column **`time_s`** (time in seconds since the start of the recording) and **one column per active channel**, named after its display name. The values are already scaled, i.e. in the unit entered.

6.2 Parquet or CSV?

	Parquet (recommended)	**CSV**
File size	small (compressed)	large (text)
Reading	very fast, also single channels	slow for large files
Opening in Excel	not directly	directly
Crash during measurement	file not readable	data up to the crash is kept

> **Note:** A Parquet file is only completed when the measurement is stopped. It cannot be opened during the measurement or after a crash. If an aborted measurement is a concern (power failure, very long measurements), use CSV.

6.3 Opening measurement data in other programs

Python (pandas):

```python
import pandas as pd
df = pd.read_parquet("TensileTest_001.parquet")
force = pd.read_parquet("TensileTest_001.parquet", columns=["time_s", "Force"])
```

MATLAB:

```
T = parquetread("TensileTest_001.parquet");
plot(T.time_s, T.Force)
```

Excel: CSV files open directly. ezDAQ writes a dot as decimal separator and commas between columns. With regional settings that use a decimal comma, import via *Data → From Text/CSV* and choose *comma* as the delimiter.

7 Analysis

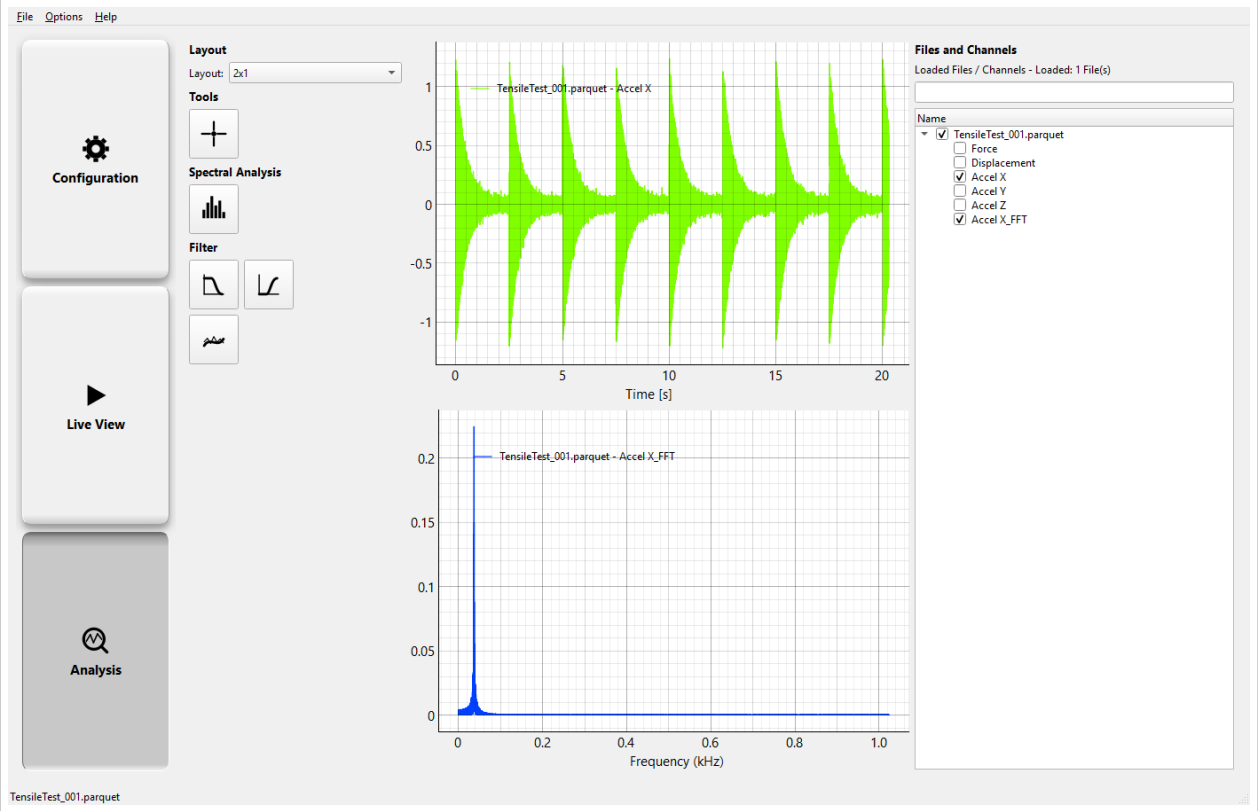


Figure 7.1: Analysis view with time signal and FFT

7.1 Loading files

Drag measurement files (`.parquet` or `.csv`) into the analysis view (drag & drop from Explorer, several at once if you like) or open them via *File* → *Load Measurement* …. The matching `_info.json` is loaded automatically. Each file appears with its channels on the right under *Files and Channels*; the search field above filters by file and channel name.

7.2 Displaying channels

- **Layout** (left): 1×1, 2×1, 3×1, 4×1 or 2×2 plots.
- **Drag a channel from the list onto a plot** to show it there. After that, the checkbox in front of the channel (or the file) shows and hides it.
- **Zoom and pan:** the mouse wheel zooms, the left mouse button pans, holding the right mouse button and dragging zooms along one axis. The small **A** button at the bottom left of the plot and *View All* in the right-click menu show everything again.
- **Readout cursor** (Tools): shows a crosshair that snaps to the nearest measured point and displays its x and y values.
- **Remove:** right-click a file or channel → *Remove File from Analysis* or *Remove Channel from Analysis* (or press **Del**). The file on disk is not changed.

7.3 Analysis functions

Function	Result	Parameter
FFT	frequency spectrum (amplitude over frequency)	–
Lowpass	filtered signal, content above the cut-off frequency damped	cut-off frequency (Hz)
Highpass	filtered signal, content below the cut-off frequency damped	cut-off frequency (Hz)
Smoothing	moving average	window size (samples)

After clicking a function, choose the channel and parameters and confirm with **Run**.

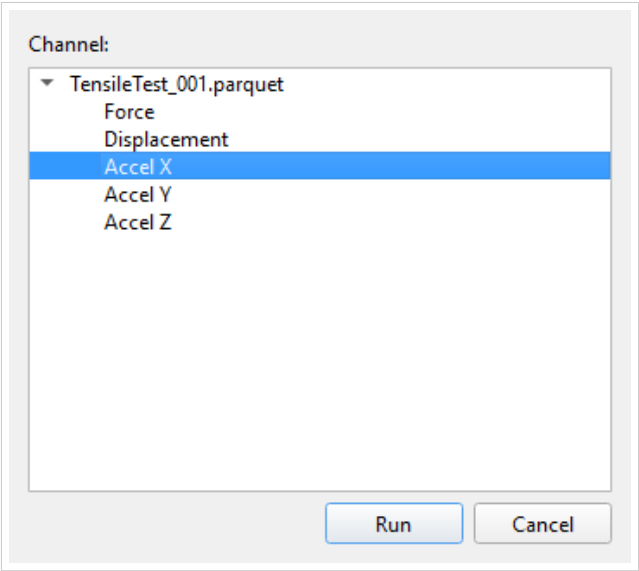


Figure 7.2: Choosing the channel for the FFT

The result appears as a **new channel below the source file**, e.g. *Accel X_FFT*, and can be dragged onto a plot like any other channel. Right-clicking it offers *Save as CSV …* or *Save as Parquet …* and *Remove Analysis Result*.

> **Note:** The FFT needs the sample rate. It is read from the metadata or, if they are missing, determined from the time column. For channels of slower modules (e.g. an NI 9210 next to an NI 9234), ezDAQ automatically uses their actual rate.

8 Settings and help

- **Language:** *Options → Language* switches between German and English, immediately and without a restart.
- **Theme:** *Options → Theme* switches between *Light* and *Dark*.
- **Manual:** *Help → User Manual (PDF)* or **F1** opens this manual in the current language.
- **Updates:** at start-up, ezDAQ checks once in the background whether a new version is available and reports it. *Help → Check for Updates …* checks by hand; the release page lists the changes and the installer.
- **About:** version and short description.

9 Troubleshooting

Problem	Cause and solution
No devices found	Is the NI-DAQmx driver installed? Is the chassis switched on and connected? A network chassis has to be reserved for this PC in NI MAX. Then click **Search Devices**.
Device *not connected*	The driver knows the device, but it does not respond. Check cable and network, test it in NI MAX.
Module *not supported*	ezDAQ does not know this module (yet); its channels cannot be selected.
Message about the sample rate at the start	The rate does not suit a module (e.g. NI 9234 only 51,200 Hz / n). Use the valid value suggested.
A measurement named … already exists	Choose another file name or switch on the number suffix.
Channel has no hardware channel yet	Assign a channel in the row via **…**, or deactivate the row.
Storage buffer turns orange/red	The target drive is too slow or almost full. Use a local drive, free up space, or lower the sample rate.
Parquet file cannot be opened	The measurement was not stopped properly (crash). Use CSV where this is a risk.
Serial trigger does not react	Check COM port and baud rate; the expected signal must match exactly. Check with **Test Serial Trigger**.
Live view stutters	Show fewer channels at once, combine channels in one plot, or shorten the time span. The measurement data is not affected.
Manual does not open	No PDF viewer is set up; the message shows where the file is.

10 Appendix

10.1 Keyboard shortcuts

Key	Function
F1	Open this manual
F5	Start recording (with storage)
F8	Stop measurement
Del	in the analysis: remove the selected file/channel from the view

10.2 Contents of the metadata file

`<name>_info.json` is a text file in JSON format. Main entries:

Entry	Content
`measurement_name`	name of the measurement
`start_time`, `end_time`, `duration_seconds`	start, end and duration
`sample_rate_hz`, `target_sample_rate_hz`	actual clock of the file and target rate set
`rate_groups`	with mixed modules: which channels were measured at which rate
`storage_format`	`parquet` or `csv`
`hardware`	detected modules with type and channel count
`channels`	all channels with name, unit, scaling, signal type and parameters